

Homework #3

Collaborators:- None

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Exercise 1.7.40.

Consider the following game: player I flips a fair coin $n+1$ times; player II flips a fair coin n times. Show that the probability that player I has more heads than player B, is equal to $\frac{1}{2}$. Is this counterintuitive, given the fact that player I flips the coin one extra time?

Solution

Since the number of coins in each pile, are different by 1, then either player 1 has more heads or more tails, and the two events are mutually exclusive. By symmetry, the probability of having more heads is the same as the probability of having more tails, and therefore the probability is half for both.

Exercise 1.7.41

Consider two events A and B, both with positive probability. When is it the case that $P(A|B) = P(B|A)$?

Solution

From Bayes update we can see that equality of the two expressions hold if, $P(A) = P(B)$.

Exercise 2.1.12

Show that the probabilities of the negative binomial distribution sum up to 1.

$$\sum_{k=r}^{\infty} \binom{k-1}{r-1} p^r (1-p)^{k-r} \quad (1)$$

$$\frac{p^r}{(r-1)!} \sum_{k=r}^{\infty} \frac{(k-1)!(1-p)^k}{(k-r)!} \quad (2)$$

But $\sum_{k=0}^{\infty} \frac{(k+r-1)!(1-p)^k}{k!} = \frac{(r-1)!}{(1-(1-p))^r} = \frac{(r-1)!}{p^r}$ because of its taylor expansion. So the sum just equals 1.

Exercise 2.7.6.

An urn contains 8 white, 4 black, and two red balls. We win 2 euro for each black ball that we draw, and lose 1 euro for each white ball that we draw. We choose three balls from the urn, and X denote our winnings. Write down the probability mass function of X .

Solution

$$P(X = -3) = \frac{8 * 7 * 6}{14 * 13 * 12} \quad (3)$$

$$P(X = -2) = 3 * \frac{2 * 7 * 8}{14 * 13 * 12} \quad (4)$$

$$P(X = -1) = 3 * \frac{2 * 1 * 8}{14 * 13 * 12} \quad (5)$$

$$P(X = 0) = 3 * \frac{4 * 8 * 7}{14 * 13 * 12} \quad (6)$$

$$P(X = 1) = 6 * \frac{4 * 8 * 2}{14 * 13 * 12} \quad (7)$$

$$P(X = 2) = 3 * \frac{1 * 2 * 8}{14 * 13 * 12} \quad (8)$$

$$P(X = 3) = 3 * \frac{8 * 4 * 3}{14 * 13 * 12} \quad (9)$$

$$P(X = 4) = 3 * \frac{2 * 4 * 3}{14 * 13 * 12} \quad (10)$$

$$P(X = 5) = 0 \quad (11)$$

$$P(X = 6) = \frac{4 * 2 * 3}{14 * 13 * 12} \quad (12)$$

$$(13)$$

Exercise 2.7.9

Suppose we throw a die twice. Let X be the product of the two outcomes.

(a) Compute the probability mass function of X .

(b) Compute $P(X = k|D)$, where D is the event that the sum of the outcomes is equal to 7.

Solution

$$p(1) = \frac{1}{36} \quad (14)$$

$$p(2) = \frac{2}{36} \quad (15)$$

$$p(3) = \frac{2}{36} \quad (16)$$

$$p(4) = \frac{3}{36} \quad (17)$$

$$p(5) = \frac{2}{36} \quad (18)$$

$$p(6) = \frac{4}{36} \quad (19)$$

$$p(8) = \frac{2}{36} \quad (20)$$

$$p(9) = \frac{1}{36} \quad (21)$$

$$p(10) = \frac{2}{36} \quad (22)$$

$$p(12) = \frac{4}{36} \quad (23)$$

$$p(15) = \frac{2}{36} \quad (24)$$

$$p(16) = \frac{1}{36} \quad (25)$$

$$p(18) = \frac{2}{36} \quad (26)$$

$$p(20) = \frac{2}{36} \quad (27)$$

$$p(24) = \frac{2}{36} \quad (28)$$

$$p(25) = \frac{1}{36} \quad (29)$$

$$p(30) = \frac{2}{36} \quad (30)$$

$$p(36) = \frac{1}{36} \quad (31)$$

$$(32)$$

For second problem.

$$P(X=6)=1/3$$

$$P(X=10)=1/3$$

$$P(X=12)=1/3$$

Exercise 2.7.10.

Let X be a random variable with probability mass function given by $p(-4)=p(4)=1/16$ and $p(2)=p(-2)=7/16$. Let $Y=aX+b$, for certain a and b .

(a) For what values of a and b does Y take values in $\{0, 1, 2, 3, 4\}$?

(b) Compute the probability mass function of Y for these choices of a and b .

Solution

(a) For Y to take values in $0,1,2,3,4$, we need to have $aX+b \in \{0,1,2,3,4\}$. Therefore the min is 0 and the max is 4, therefore the values of a and b are $1/2$ and 2 (I got it by just setting 0 to $-4X+b$ and figuring out the coefficients), respectively, it could also be $-1/2$ and 2 because it symmetric with respect to the origin.

(b) Since Y is linear and therefore an injective function, we can just map the pmf of X to its values in Y . Therefore the pmf of Y is given by:

$$p(0) = 1/16$$

$$p(1) = 7/16$$

$$p(2) = 0$$

$$p(3) = 7/16$$

$$p(4) = 1/16$$

Exercise 2.7.12.

We throw a fair die twice. Let X be the sum of the outcomes, and let Y be the highest outcome that we see. Finally, A is the event that the outcome of the first throw is higher than the outcome of the second.

- (a) Compute the probability mass function of Y and compute $E(Y)$.
- (b) Are X and Y independent?
- (c) Are $\{X = 5\}$ and A independent events?

Solution

$$p(1) = 1/36$$

$$p(2) = 3/36$$

$$p(3) = 5/36$$

$$p(4) = 7/36$$

$$p(5) = 9/36$$

$$p(6) = 11/36$$

$$E(Y) = \sum_{k=1}^6 k * p(k) = 158/36$$

For question b, just from common sense if the sum is 6, the highest value can't be 2. Therefore $P(X = 6, Y = 2) \neq P(X = 6) * P(Y = 2)$, and therefore they are not independent.

$\{X=5\}$ and A aren't independent, because $P(A|X = 5) = 1/2 \neq P(A) = 15/36$. Therefore they aren't independent.